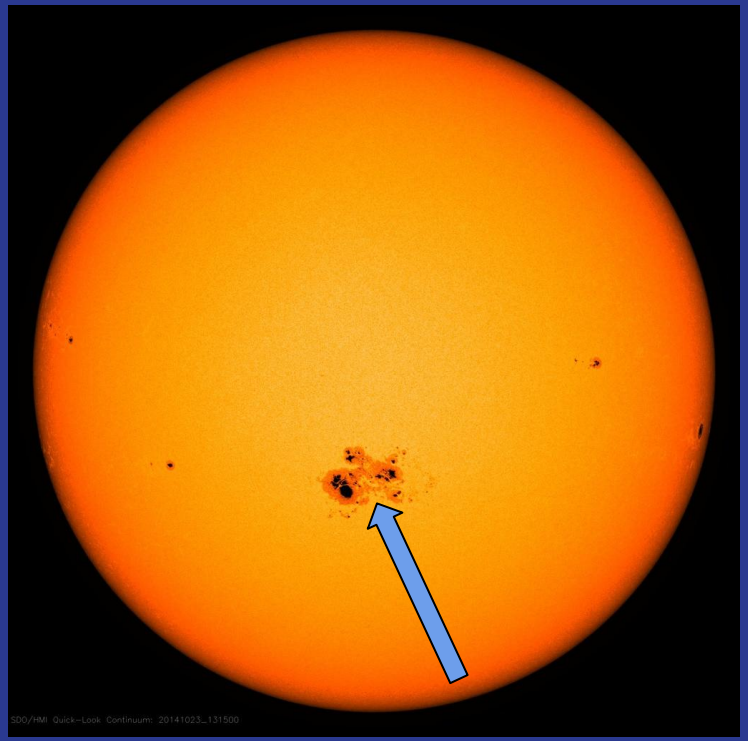
Exploring the activity levels of relatively quiet main-sequence stars

Context

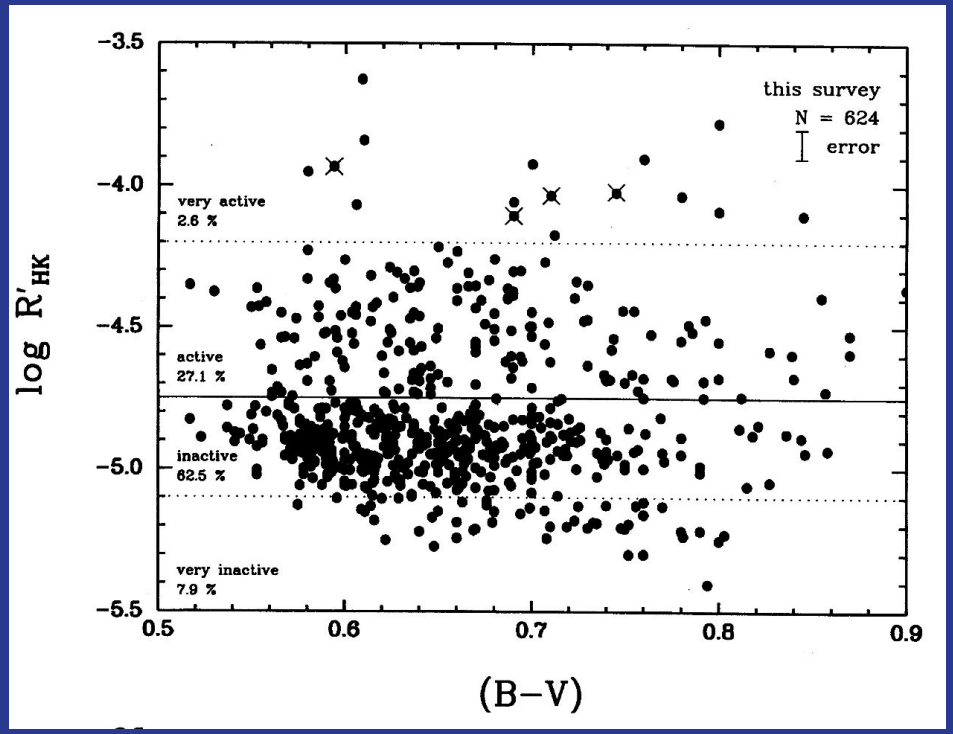


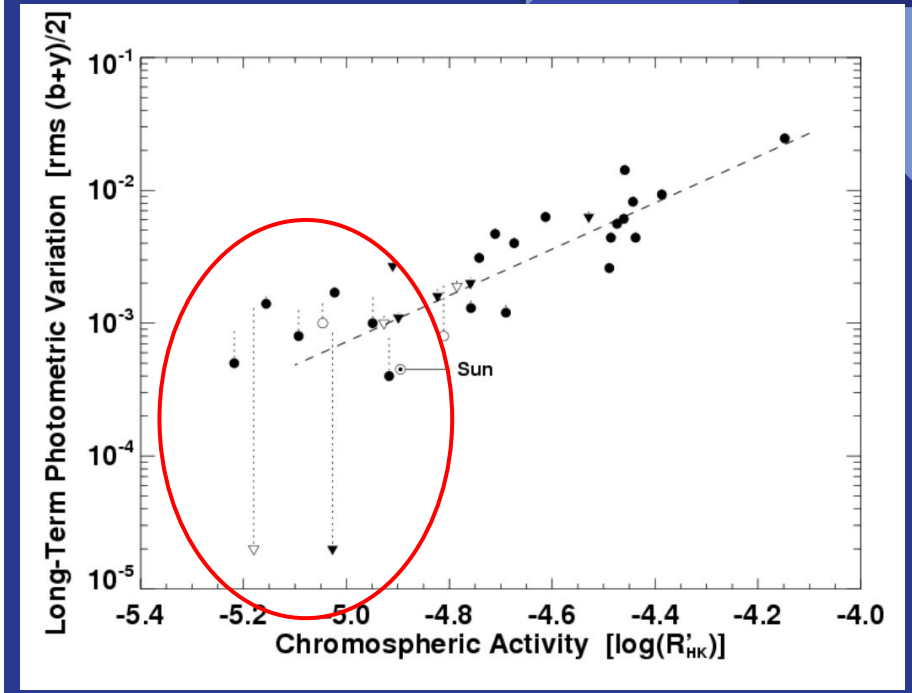
Stellar activity can perturb the RV and transit methods, hiding signals of exoplanets.

- Ca II H & K fluxes are a good proxy for activity level.
- Used to identify non-variable stars.

Wilson 1978, Noyes et al. 1984, Baliunas et al. 1995, ...

Context

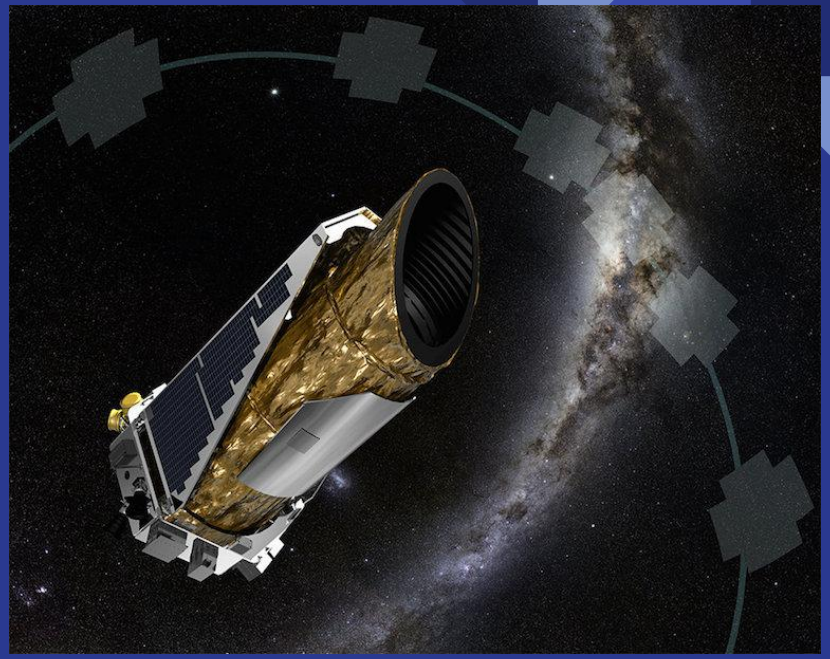




Henry et al. 1996

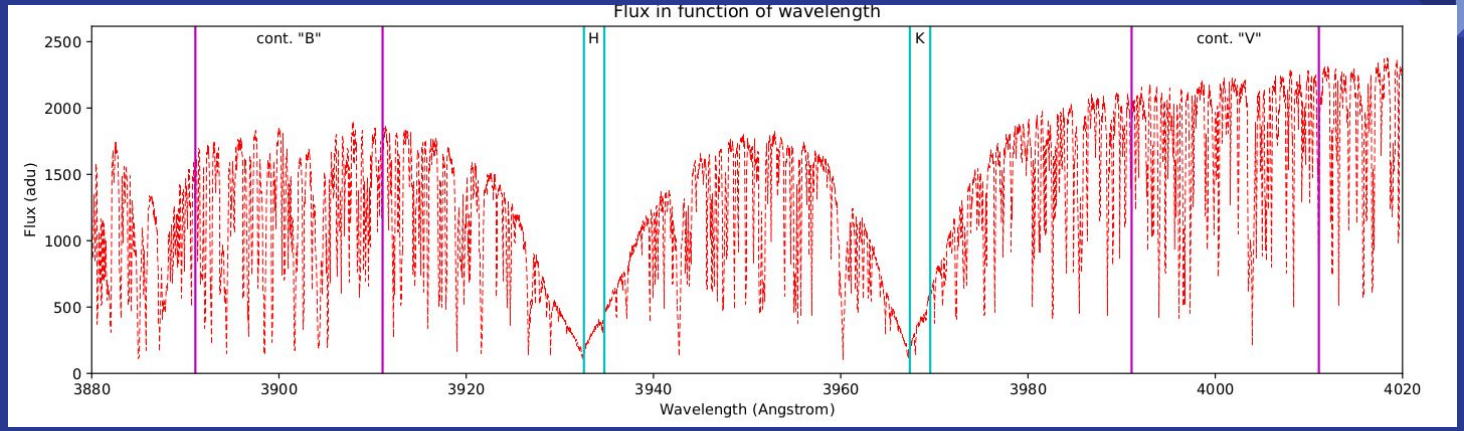
Lockwood et al. 2007

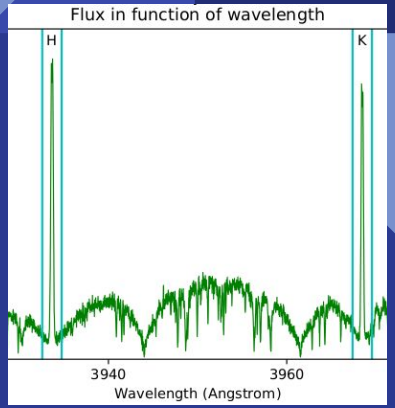
Data



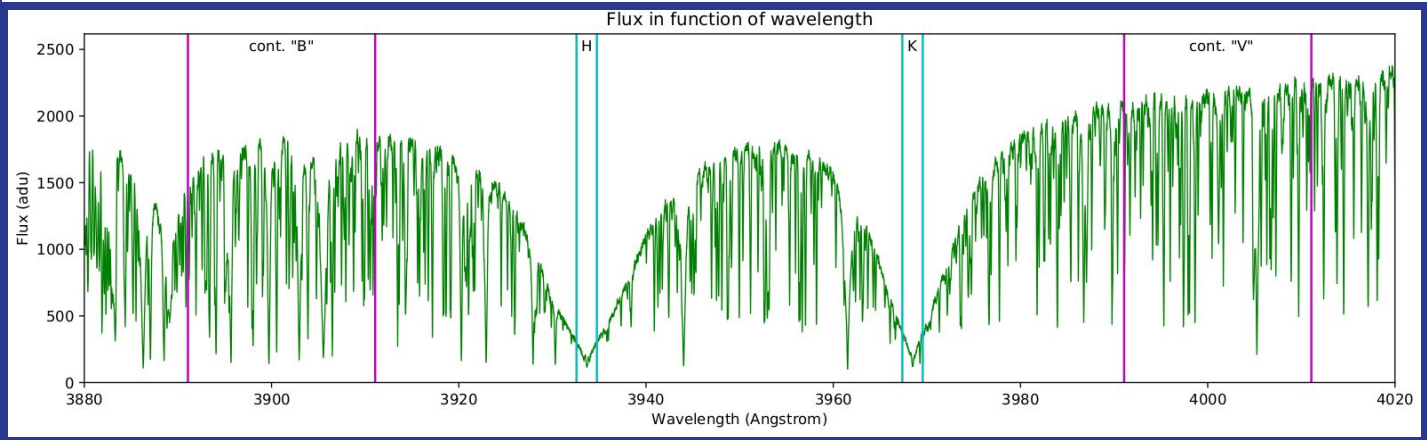
Sample of 80 stars ranging from F to M, observed simultaneously by UVES and Kepler/K2

Radial velocity correction





EPIC 211529603

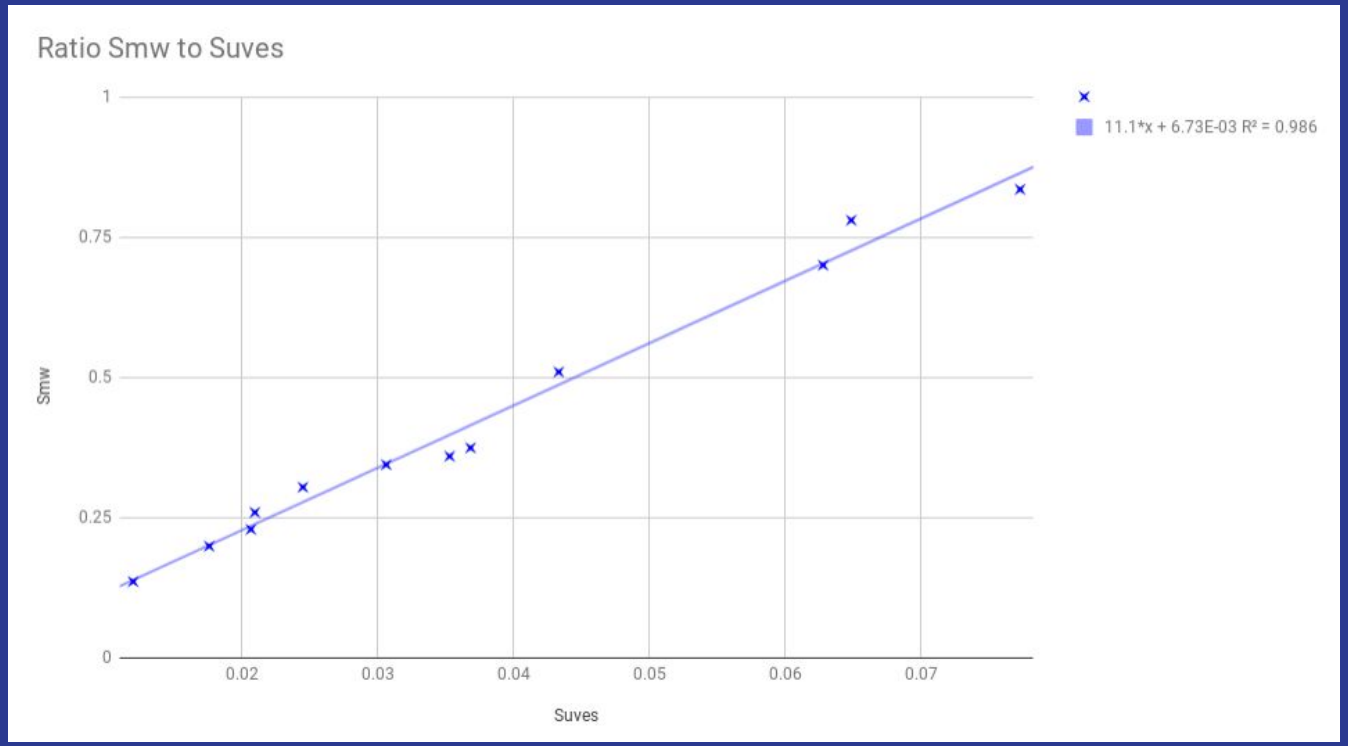


$$S = \frac{H + K}{B + V}$$

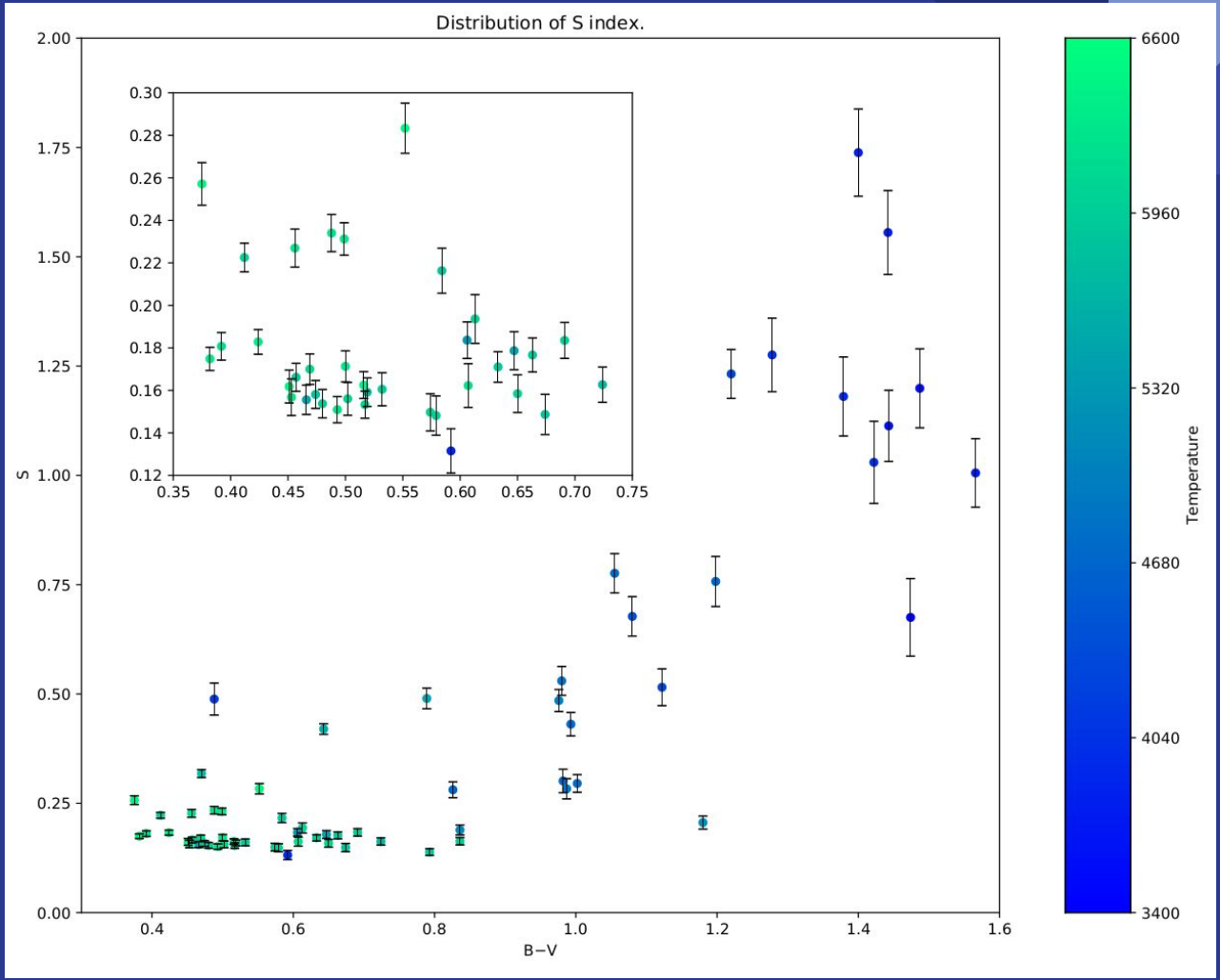
EPIC 211842729

S index calibration

Done with 12 reference stars



S index


Distribution of S index.
S
B−V
Temperature
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0.4
0.6
0.8
1.0
1.2
1.4
1.6
6600
5960
5320
4680
4040
3400
0.30
0.28
0.26
0.24
0.22
0.20
0.18
0.16
0.14
0.12
0.35
0.40
0.45
0.50
0.55
0.60
0.65
0.70
0.75

log(R'hk)

Middelkoop 1982,
Noyes et al. 1984

$$R'_{HK} = R_{HK} - R_{phot}$$

$$\log(R_{phot}) = -4.898 + 1.918 \cdot (B-V)^2 - 2.893 \cdot (B-V)^3$$

$$R_{HK} = 1.34 \cdot 10^{-4} \cdot C_{cf} \cdot S$$

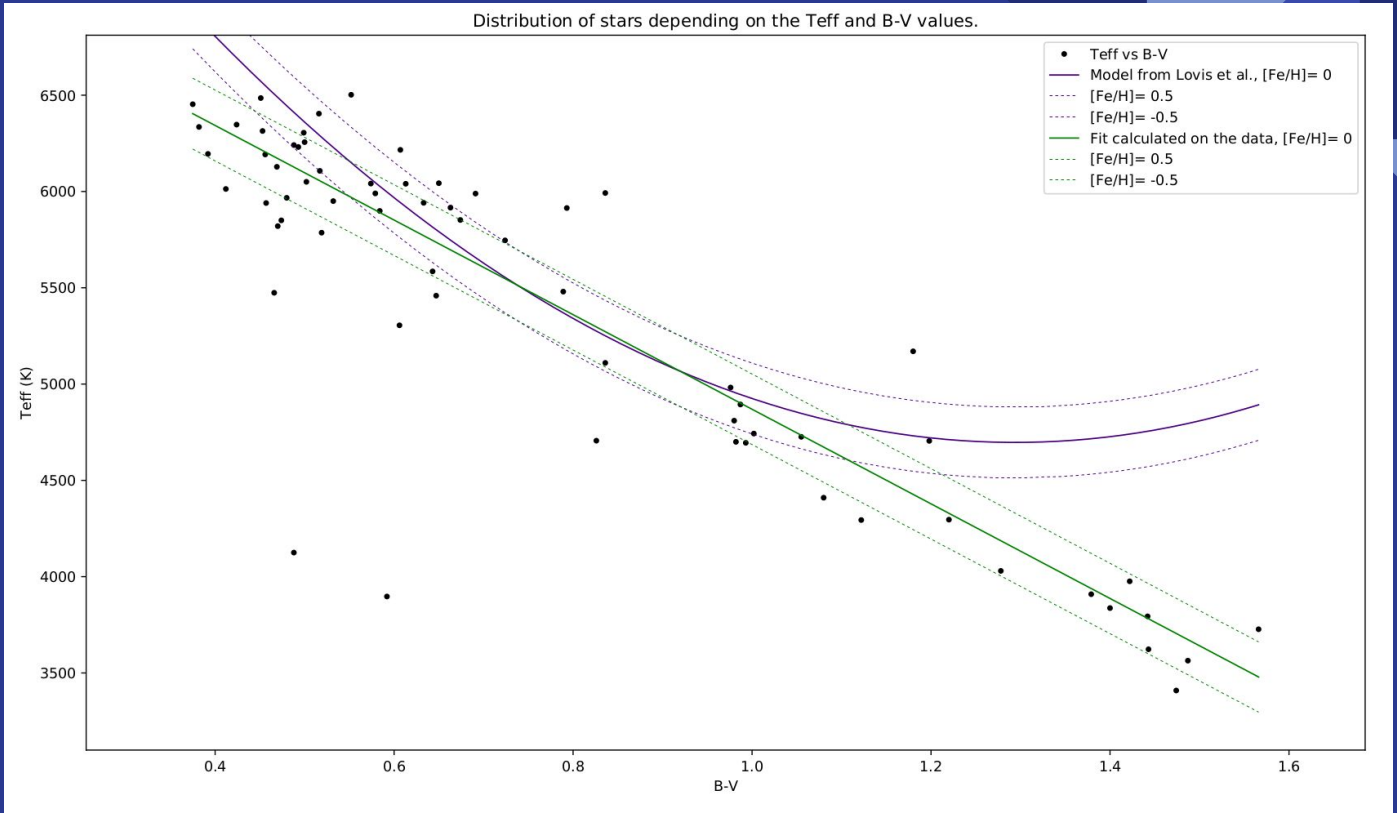
$$C_{cf}(B-V) = 1.13 \cdot (B-V)^3 - 3.91 \cdot (B-V)^2 + 2.84 \cdot (B-V) - 0.47$$

$$R'_{HK,new} = R'_{HK} \cdot \left(1 + \frac{368 \cdot \left[\frac{Fe}{H}\right]}{9114 - 6827 \cdot (B-V) + 2638 \cdot (B-V)^2}\right)^4$$

$$T_{eff} = 9114 - 6827 \cdot (B-V) + 2638 \cdot (B-V)^2 + 368 \cdot \left[\frac{Fe}{H}\right]$$

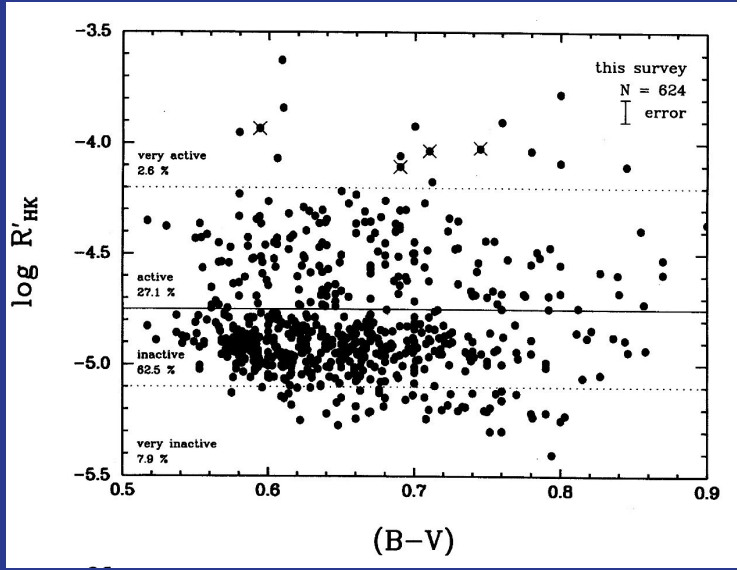
Sousa et al. 2008,
Lovis et al. 2011

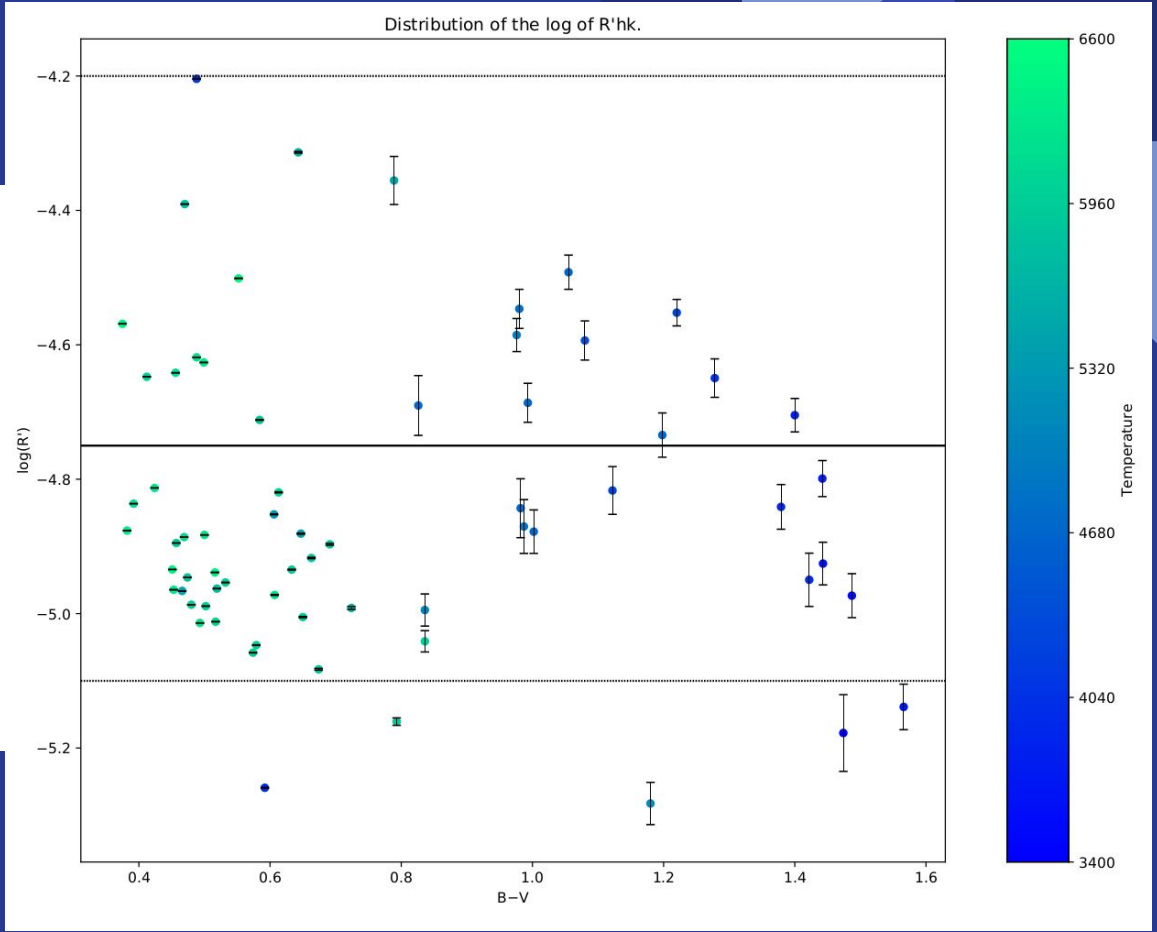
log(R'hk)



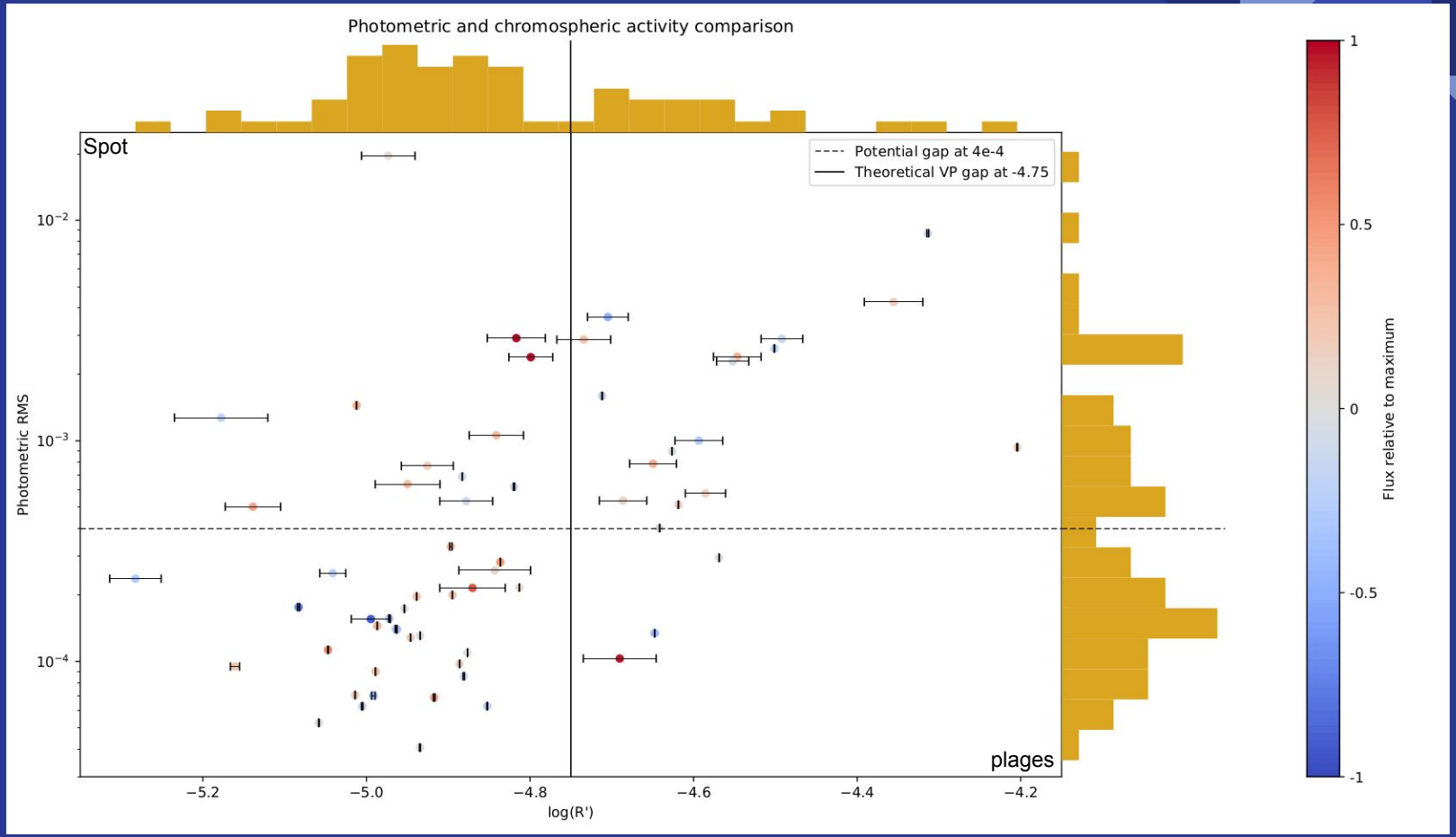
$$T_{eff} = 7325 - 2456 \cdot (B - V) + 368 \cdot \left[\frac{Fe}{H} \right]$$

log(R'hk)


this survey
N = 624
error
very active
2.6 %
active
27.1 %
inactive
62.5 %
very inactive
7.9 %
log R'HK
(B−V)
−3.5
−4.0
−4.5
−5.0
−5.5
0.5
0.6
0.7
0.8
0.9


Distribution of the log of R'hk.
log(R')
B−V
Temperature
−4.2
−4.4
−4.6
−4.8
−5.0
−5.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
6600
5960
5320
4680
4040
3400

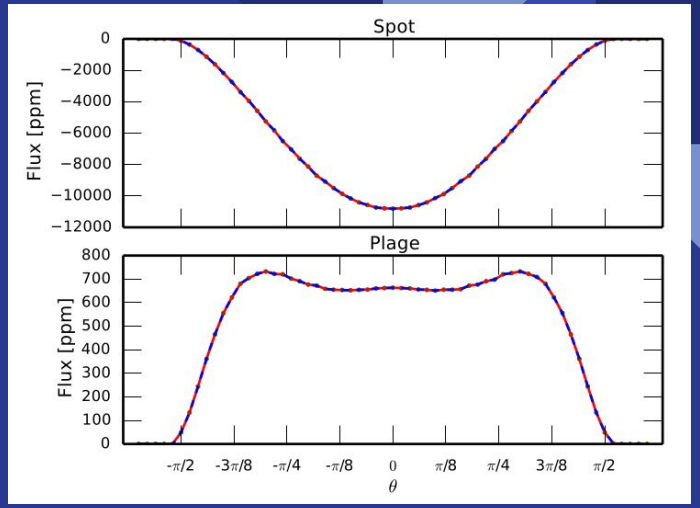
Photometric comparison



Conclusion

- There seems to be two populations in variability

- We note the presence of two regions:
  - High RMS / low log(R'hk)
  - Low RMS / high log(R'hk)

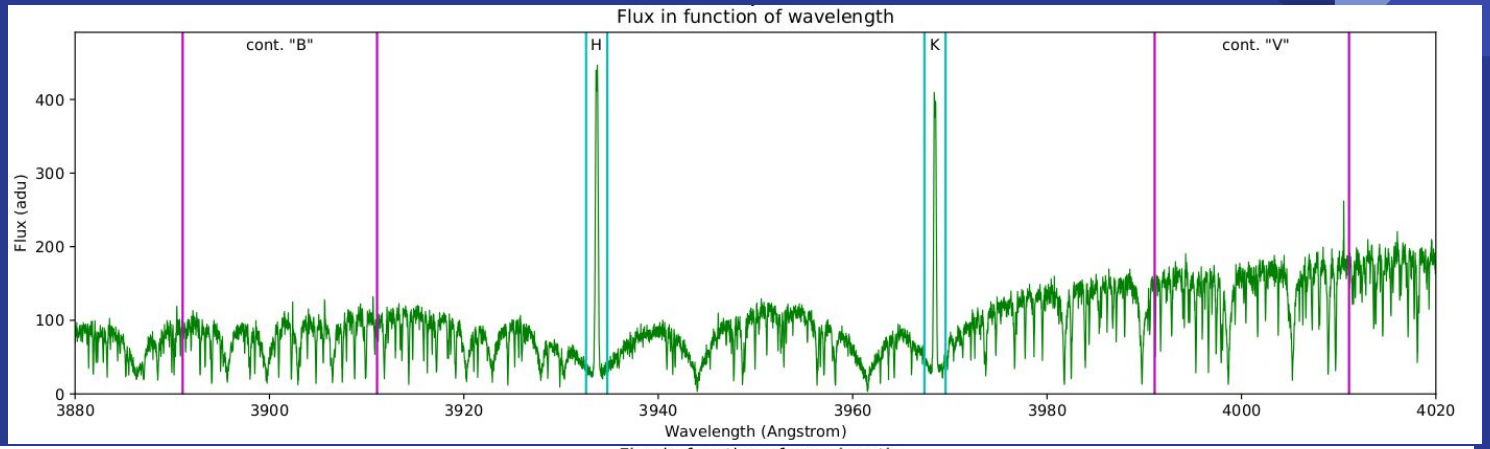


Dumusque et al. 2014

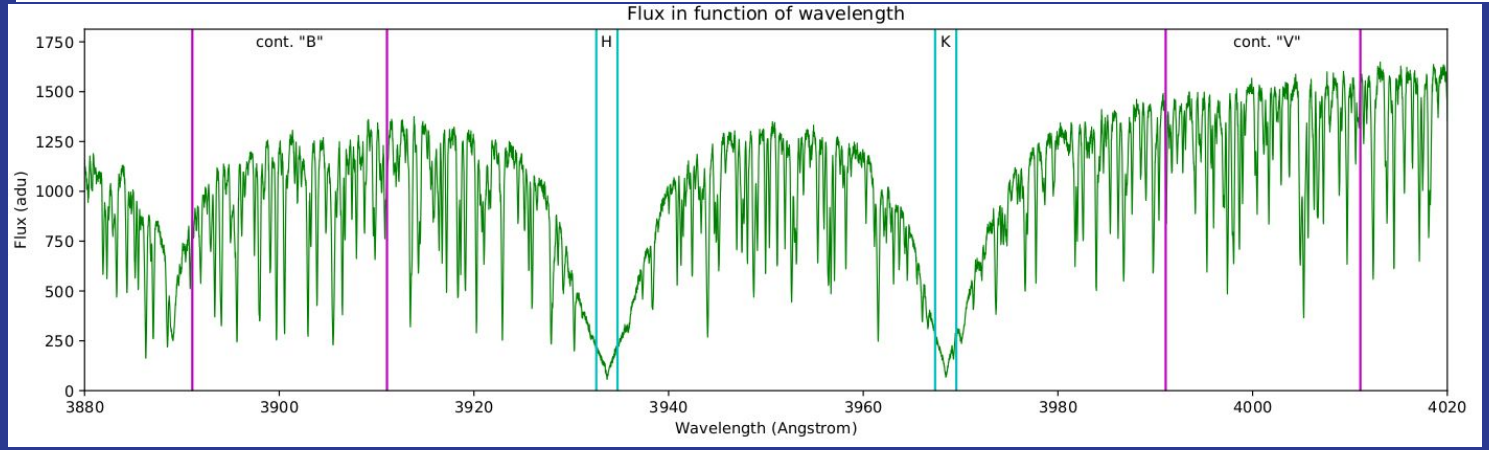
This implies that the linear trend observed previously isn't maintained for log(R'hk)<-4,6 .

Annex

HK emission



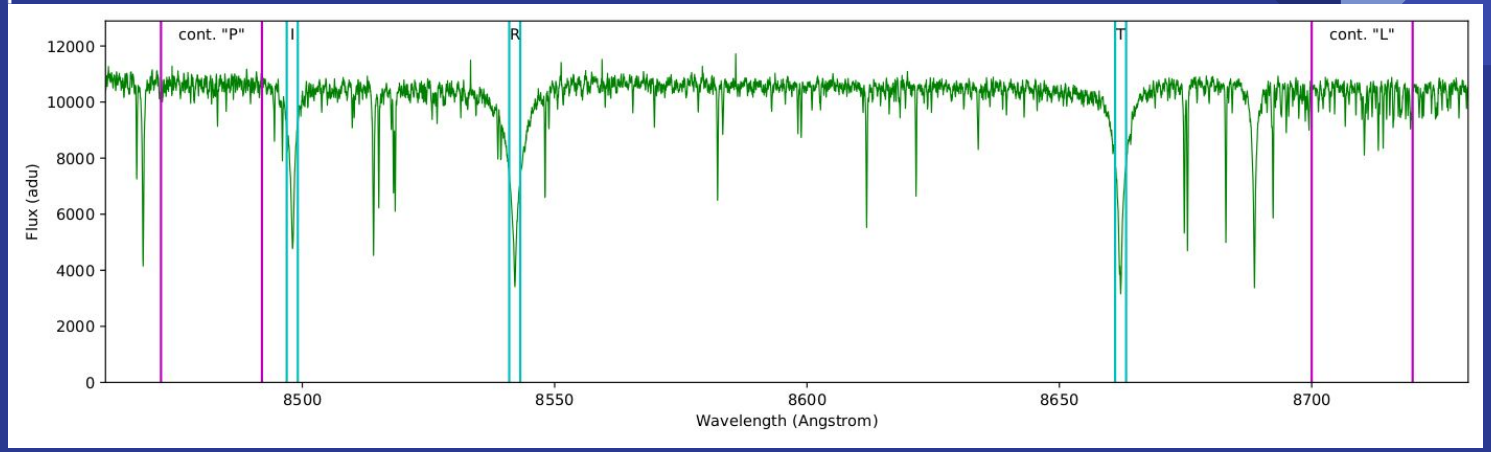
EPIC 211529603
“active”



EPIC 211516730
“inactive”

Annex

IRT comparison



EPIC 211529603
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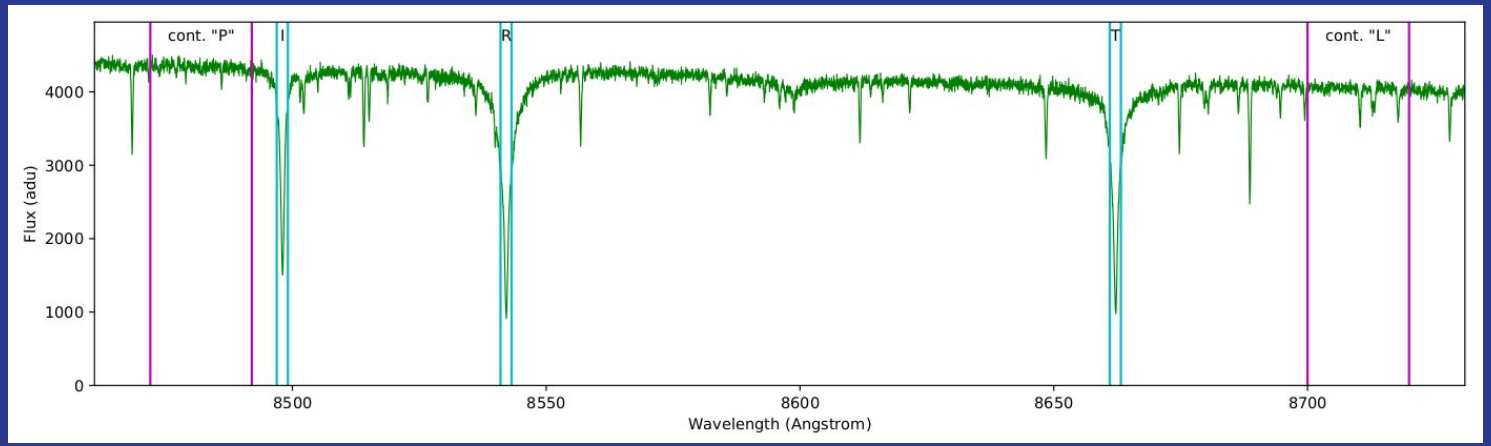

EPIC 211516730
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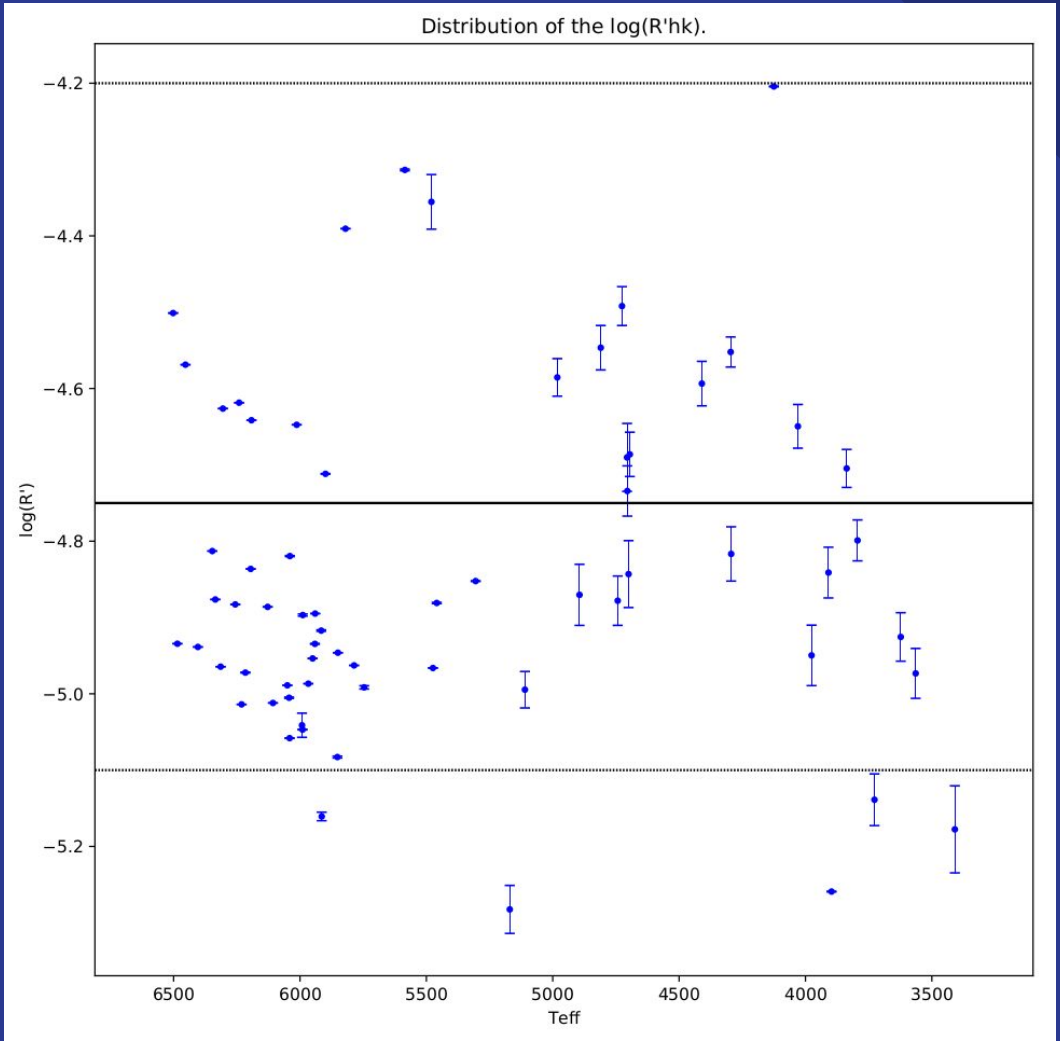
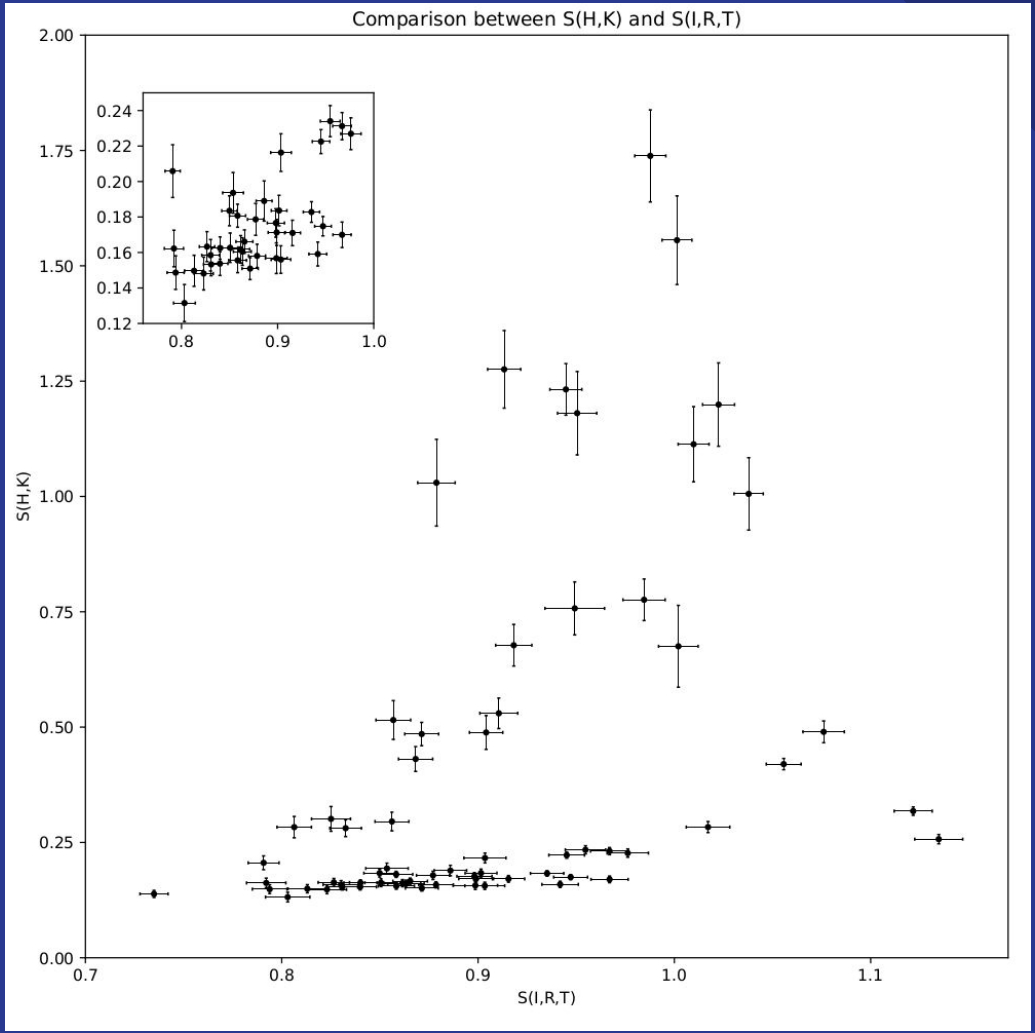
Annex

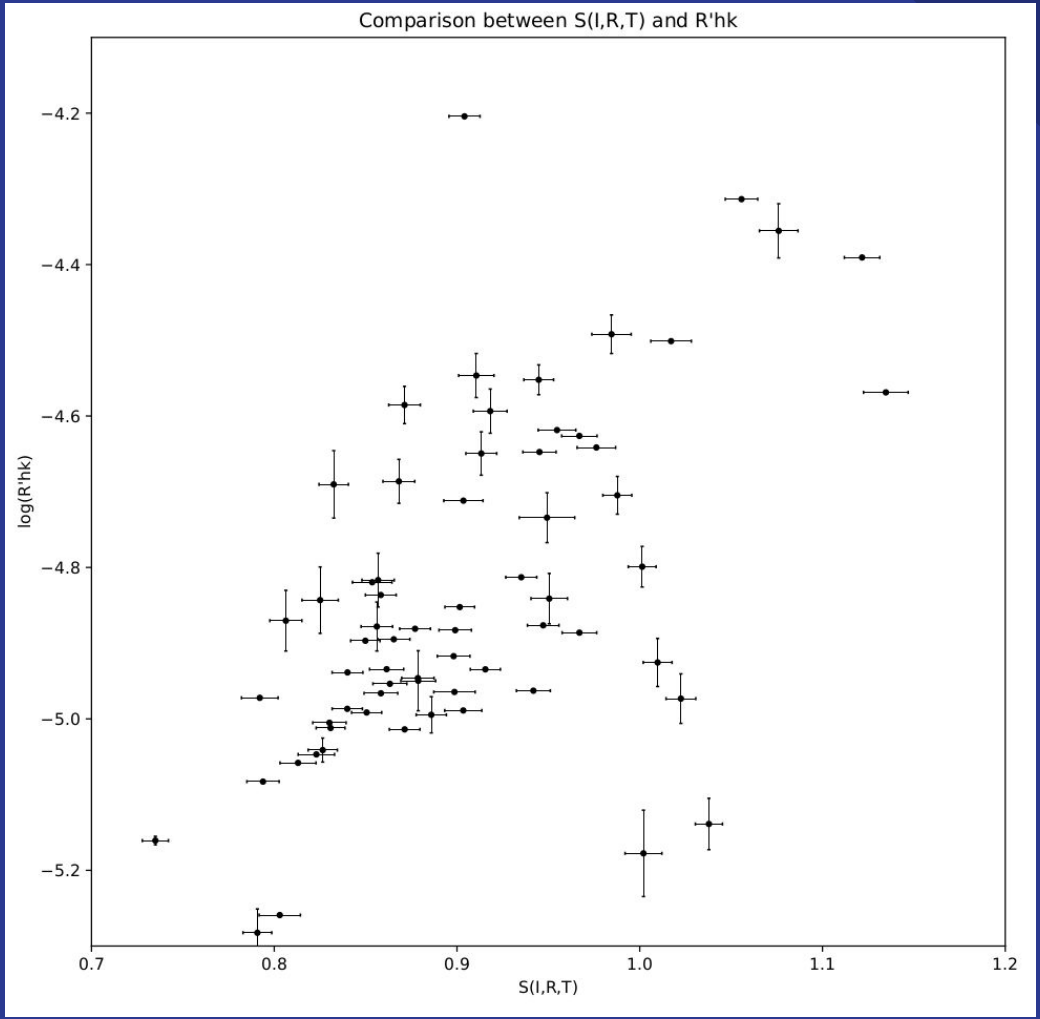
R vs Teff



Annex

S(H,K) vs S(I,R,T)



Annex

log(R’hk) vs S(I,R,T)